

Beyond Monolithic Libyan Arabic: A Sub-Regional Speech Corpus and Dialect Identification Benchmark

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Abstract

Spoken Arabic dialect identification (DID) has advanced quickly, but Libyan Arabic remains under-served and, where it appears, is labeled only at the country level, hiding the variation between the Libyan coast and interior. We present a Libyan Arabic speech corpus annotated at the sub-regional level: Tripolitania (west), Cyrenaica (east), and Fezzan (south). It contains 133.7 hours of single-speaker segments from 645 speakers across 161 channels (YouTube and TikTok sources). An automated pipeline (diarization, speaker embedding, clustering, cross-episode linking) produces silver labels from channel provenance. One annotator then reviewed the resulting speaker clusters and confirmed the provenance region for 95.5% of the kept speakers (a within-annotator figure). We release speaker-disjoint splits. An off-the-shelf Arabic DID model recognizes our verified Libyan speech as Libyan only 74% of the time and cannot resolve the region. On the new three-way task, the encoder matters more than the classifier: a probe over speaker-recognition ECAPA embeddings reaches 0.525 macro-F1, over a language-identification ECAPA reaches 0.660, and fine-tuning that encoder reaches 0.702 (75.5% accuracy), far above the 0.20 floor. A channel-disjoint re-test reaches 0.609, versus 0.702 speaker-disjoint; Fezzan is hardest at 0.313. The task is learnable, but Fezzan performance remains low. We release the corpus, cached embeddings, and code.¹

1 Introduction

Arabic is a collection of spoken varieties that differ enough to degrade speech systems trained on another variety. Dialect identification (DID) predicts which variety a recording is in, and it has become a standard pre-processing step for Arabic ASR and a

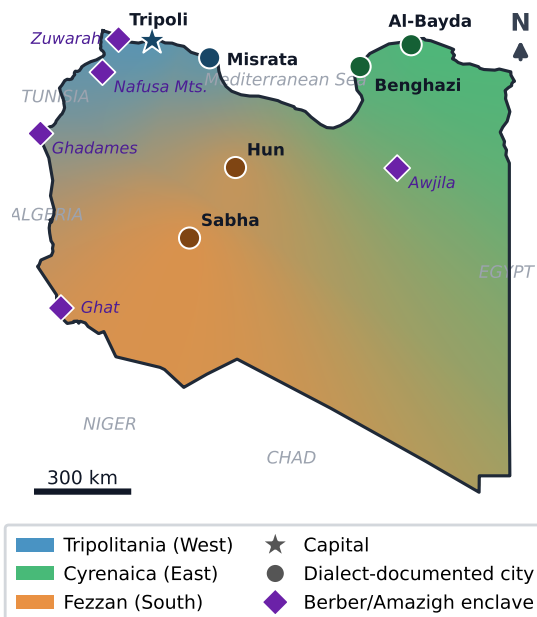


Figure 1: The three Libyan Arabic macro-dialect regions used as labels: Tripolitania (west), Cyrenaica (east), and Fezzan (south). The fill is a soft inverse-distance blend of each region’s dialect-documented cities (stars and dots), not a hard border, reflecting the gradual transitions reported in the literature (Pereira, 2010; Benkato, 2014). Diamonds mark Berber/Amazigh enclaves, which lie outside the Arabic continuum.

benchmark task in its own right (Shon et al., 2020; Talafha et al., 2025). The field has converged on a small set of broad labels, usually one per country or one per region group.

Libyan Arabic has received little of this attention. Large multi-dialect collections such as ADI-17 and ADI-20 (Shon et al., 2020; Elleuch et al., 2025) represent it only as a single country-level class, and no prior resource labels it below that level. On our human-verified data, an off-the-shelf Arabic dialect classifier labels only 74% of playlist-level speaker clusters as Libyan and scatters the rest across Egyptian, Algerian, and Levantine labels (Section 5).

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¹Released at <https://huggingface.co/datasets/NourZohdy/libyan-subregional-did>; archived at <https://doi.org/10.5281/zenodo.22802659>.

Consistent with known cross-corpus degradation in Arabic DID (Shon et al., 2020; Sullivan et al., 2023), country-level tools do not transfer to these data or resolve their regions.

A second problem is the label itself. Treating “Libyan” as one dialect ignores variation that speakers themselves notice. The country splits into three historical regions with different settlement histories and contact languages: Tripolitania in the west, Cyrenaica in the east, and Fezzan in the south (Figure 1) (Pereira, 2010; Benkato, 2014; Benkato and Pereira, 2016). These regions differ in lexicon and phonology, yet no speech resource labels Libyan data at this level. A model trained on country-level labels cannot use this variation, and a dataset without regional labels cannot show whether the variation is recoverable from audio.

We address both gaps with a corpus and a benchmark. Our contributions are:

- **A sub-regional Libyan Arabic speech corpus.** 133.7 hours of validated single-speaker segments from 645 speakers and 161 channels, each labeled Tripolitania, Cyrenaica, or Fezzan (Section 3).
- **A partition and validation methodology.** A reproducible pipeline that turns raw broadcast and social-media audio into speaker-disjoint, region-labeled segments. It pairs provenance-based regional tagging with a human review protocol for checking and correcting those initial labels: channel-normalized merge bands, an evidence floor, and a risk-ranked review whose decisions confirm the provenance region 95.5% of the time (within-annotator consistency; Sections 3–4).
- **A three-way DID benchmark.** Speaker-disjoint splits and a set of models, from a majority floor and an off-the-shelf classifier through frozen-embedding probes to an end-to-end fine-tuned encoder, with per-class results and an analysis of which regions are separable. The strongest model, a fine-tuned language-identification ECAPA, reaches 0.702 macro-F1 against a 0.20 floor (0.609 under a channel-disjoint split); Fezzan is the hardest class (Section 5).

The benchmark also tests the labels. Our regional labels start from the channel’s known origin, not from per-utterance perceptual annotation. If

a classifier trained on these labels generalizes to unseen speakers above chance, the labels carry a learnable regional signal. Section 5 shows that they do, although part of that signal is channel-correlated (Section 5.7).

2 Related Work

Arabic dialect identification corpora. Most spoken Arabic DID work builds on broadcast collections. The MGB challenges released Egyptian and Moroccan broadcast speech (Ali et al., 2017, 2019), and ADI-17 turned YouTube audio into a 17-way country-level dialect task (Shon et al., 2020). ADI-20 extended this direction to twenty varieties (Elleuch et al., 2025), and QASR (Mubarak et al., 2021) and MASC (Al-Fetyani et al., 2023) added large transcribed Arabic broadcast corpora. The NADI shared tasks have run for several years and cover both text and, more recently, spoken dialect identification at the country level (Talafha et al., 2025). Casablanca (Talafha et al., 2024) and CAFE (Lachemat et al., 2024) are recent multi-dialect speech resources. None of these label data at the sub-national level for Libya.

Libyan Arabic in prior work. Libyan appears as one class in country-level collections such as ADI-20 (Elleuch et al., 2025) and in recent Libyan-focused studies (Essgaer et al., 2025), but always as a single label. To our knowledge no prior speech resource separates Tripolitania, Cyrenaica, and Fezzan. Our corpus extends the country-level Libyan class of ADI-20 with labels for three regions inside Libya. Casablanca also sources dialogue from television drama and contains code-switching, which supports our choice of sources and our quality gates. It covers eight national dialects and no Libyan.

Models for spoken DID. Early systems used i-vectors and x-vectors with a back-end classifier (Snyder et al., 2018). Strong speaker and language embeddings such as ECAPA-TDNN (Desplanques et al., 2020) and the VoxLingua107 language-ID model (Valk and Alumäe, 2021) are now common encoders, and self-supervised models like wav2vec 2.0 (Baeovski et al., 2020), XLS-R (Babu et al., 2022), and HuBERT (Hsu et al., 2021) provide transferable features for low-resource dialect tasks (Sullivan et al., 2023). We compare two ECAPA encoders—a speaker-recognition model and the VoxLingua107 language-identification model (Valk and Alumäe, 2021), which share an

architecture but differ in pretraining objective—alongside a frozen self-supervised encoder, and we fine-tune the language-identification ECAPA end to end (Section 5). Table 1 places our corpus against prior resources.

3 The Corpus

3.1 Sources and regions

The audio comes from public Libyan media: television drama, talk shows, and podcasts on YouTube, plus short clips from TikTok. A *source* is one YouTube channel or playlist, or one TikTok account; the paper calls this unit a *channel* throughout. A *recording* is one video or clip from a source. The released corpus has 161 sources: 57 on YouTube and 104 on TikTok. TikTok supplies 15.2 of 133.7 hours (11%) but 65% of the sources. We included its short clips to obtain informal speech from more creators; we expected YouTube’s scripted and broadcast material to use a more formal register. TikTok also supplied additional Fezzan data after the available broadcast sources ran out. We group every source into one of three regions using its known origin. The channel’s region serves only as an initial, provenance-based label for each segment. Every candidate speaker that passed the evidence floor (Section 3.3) was reviewed and could be confirmed, corrected, or excluded; released labels are the three broad regions rather than city-level or personal dialect labels. The same channel-provenance assumption underlies ADI-17 and similar broadcast DID corpora (Shon et al., 2020).

3.2 Automated pipeline

Each channel is processed independently, in four stages.

Segmentation. We run speaker diarization (Bredin et al., 2020) to split each episode into single-speaker turns, then cut long turns at low-energy pauses to keep segments between three and ten seconds. Without hand-aligned speaker timestamps, we report no diarization error rate. The quality gates below and human speaker review reduce segmentation errors but do not measure them.

Embedding and quality gates. Every segment gets a 192-dimensional ECAPA-TDNN speaker embedding (Desplanques et al., 2020; Ravanelli et al., 2021). We then drop non-speech and low-quality segments with three gates. The music gate is deliberately a conjunction: a segment is dropped

only if a PANNs audio tagger (Kong et al., 2020) gives it a high music score *and* a low speech score. Libyan drama often has speech over a music bed, and this speech is valid data. On the playlist used to calibrate the filter, a music-only rule wrongly flagged 27% of the segments. The conjunction reduced the flags from 681 to 8. We listened to these eight segments and confirmed that they were correct drops. This check covers one playlist; it is not a precision or recall estimate for the music gate. A voice-activity speech-ratio floor (Silero Team, 2024) drops applause and silence-padded turns. A signal-to-noise floor drops unusable noise.

Clustering and linking. Within each channel (one playlist or source), agglomerative clustering groups segment embeddings into speakers after a median absolute deviation (MAD) audit removes off-centroid segments. Drama playlists record all speakers in one studio, which inflates every pairwise cosine similarity between speaker centroids; a fixed merge threshold over-merges. A pair must instead be an outlier against *that playlist’s own* similarity distribution, with a floor of $\max(0.60, \text{median} + 4 \cdot 1.4826 \cdot \text{MAD})$. A fixed threshold queued more than 600 pairs for review on the largest playlists alone; the adaptive band queued 75 across all playlists. We then compare centroids across episodes and channels; automatic links stay within one region, so a recurring host becomes one identity. A confident cross-region match is never merged automatically: it means one of the two region labels is wrong, and only a human resolves it.

Triage scoring. For every candidate speaker we compute a risk score from an off-the-shelf HuBERT Arabic dialect classifier,² the VoxLingua107 language-ID model (Valk and Alumäe, 2021), embedding cohesion, a music fraction, and a predicted-gender mix. The risk score only orders the human review queue, hardest cases first; it rejects and relabels nothing.

3.3 Human validation

The automated output is silver-quality. Before review, an evidence floor removed very small clusters, those with under 45 s of speech or fewer than five segments. This removed 1,300 of 2,069 auto-

²IbrahimAmin/hubert-arabic-spoken-dialect-classifier on the Hugging Face Hub (Amin, 2025): a HuBERT model (Hsu et al., 2021) fine-tuned on QASR and ADI-17 to label Modern Standard Arabic and 17 regional dialects, with a single Libyan (LIB) class.

Resource	Source	Granularity	Libyan	Sub-national
ADI-17 (Shon et al., 2020)	YouTube	17 countries	yes	no
ADI-20 (Elleuch et al., 2025)	broadcast/YT	20 varieties	yes	no
NADI 2025 (Talafta et al., 2025)	TV series	8 countries	no	no
Casablanca (Talafta et al., 2024)	TV	8 dialects	no	no
MGB-3/5 (Ali et al., 2017, 2019)	broadcast	EGY / MOR	no	no
This work	broadcast/social	3 Libyan regions	yes	yes

Table 1: Our corpus against prior Arabic spoken-dialect resources. Prior resources that include Libyan label it only at the country level.

matically linked speaker identities, containing only 3.8% of the speech (5.4 hours). The 769 above-floor speakers remained. One annotator—an author raised in Libya and familiar with its regional dialects—reviewed each candidate speaker cluster in a custom interface. The review screen showed the channel name, provenance region, proposed label, tier, and triage risk score. It showed 3–10 clips from the start, middle, and end of each speaker’s timeline, preferring clean, long ones.

Each reviewed speaker received one decision: *verified*, *wrong_region* (with a corrected region), *non_libyan*, *msa*, *bad_audio*, *mixed_cluster*, or *unsure*. Of 769 reviewed speakers, we excluded 76 (34 *msa*, 28 *bad_audio*, 13 *non_libyan*, and 1 *mixed_cluster*) and kept 693 (90.1%). Of those kept speakers, 31 were Libyan but *wrong_region* and were relabelled. Table 9 (Appendix A) separates the counting units used in validation. The 693 kept speakers correspond to 799 playlist-level speaker clusters before cross-playlist merging, which is the unit used in Section 5.1 and Figure 4. After manual pair matching and merging across playlists, 645 final speakers remained. Thus, 799, 693, and 645 describe the same kept speakers in different counting units, while 769 is the earlier count before the 76 exclusions.

The annotator saw the proposed label and channel region before each decision, which may have anchored them. The classifier in Section 5.1 fed the triage score, so its evaluation partly uses labels it helped shape.

The channel-provenance region was confirmed for 95.5% of kept speakers (within-annotator consistency); of the 31 corrections, 20 moved from Tripolitania to Cyrenaica, 9 from Cyrenaica to Tripolitania, 1 from Tripolitania to Fezzan, and 1 from Fezzan to Cyrenaica. The risk score ranked rejected speakers ahead of kept ones (AUC 0.82, Figure 2). Automatic MSA probability flagged only 12 of 769 speakers above 0.5, while human

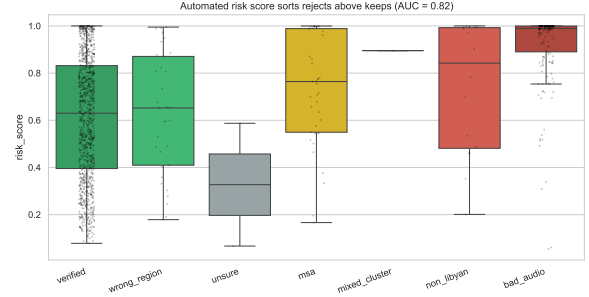


Figure 2: Triage risk score vs. human decisions. Higher risk concentrates among rejected speakers (AUC 0.82).

review removed 34 MSA and 13 non-Libyan speakers. The kept corpus is therefore mainly dialectal rather than read speech (speech read from a prepared text, usually news-style MSA with little regional dialect), although the segment audit below shows that some MSA remains at segment level.

Blind second-annotator review. A second Libyan-raised annotator, also a co-author, re-judged all 645 released speakers blind to provenance, channel, and triage information. He confirmed 525 labels (81.4%, $\kappa = 0.72$); Table 2 gives results by region, without per-region κ because the released label is constant within a region. This is lower than the 95.5% within-annotator rate. Two annotators do not always agree on the region of a speaker from short clips. We therefore call the released labels reviewed provenance labels rather than gold labels. The benchmark measures how well a model reproduces these reviewed labels; the disagreement limits how far they can be read as perceptual dialect judgments.

Segment-level purity audit. The first annotator reviewed 207 test segments, 69 per region, in random order and without seeing the released label (Table 3). Because the first annotator performed this audit, its region field measures segment purity and within-annotator consistency, not inter-annotator agreement. Code-switching and mixed-speaker au-

Released region	<i>n</i>	Confirmed	Trip.	Cyr.	Fez.	Other
Tripolitania	275	77.8%	214	40	16	5
Cyrenaica	240	82.1%	11	197	22	10
Fezzan	130	87.7%	6	8	114	2
All	645	81.4%	231	245	152	17

Table 2: Blind second-annotator review by released region: speakers he judged as Tripolitania, Cyrenaica, Fezzan, or other (13 MSA, 3 non-Libyan, 1 mixed).

Property	Trip.	Cyr.	Fez.	Weighted
MSA	13.0	15.9	33.3	17.3
Code-switching	0.0	1.4	0.0	0.6
Mixed speakers	0.0	0.0	0.0	0.0
Indeterminate	1.4	2.9	1.4	2.1
Different region	14.5	14.5	26.1	16.2
Cannot tell	17.4	23.2	49.3	24.7

Table 3: Segment-level purity audit (%) on the released test split: 23 segments per region $\times$ duration band, at most three per speaker; weighted rates use test-split region shares. Wilson 95% intervals are in the released CSV.

dio were rare, whereas MSA and uncertainty about region were common. Fezzan stood out, with an MSA rate more than double that of the coastal regions. A 3–10 s clip was often too short to place a speaker, with 37.7% of short segments marked *cannot tell*. The low code-switching count, 1 of 207 segments, does not mean that the corpus has no code-switching. This audit used one listener judging short clips, so rare events are not well measured; a larger segment-level audit would be needed to bound them.

3.4 Statistics

Table 4 reports the corpus after validation; it contains 78,411 segments with a median segment length of 6.0 seconds. Fezzan is smaller than the other regions, which reflects how little southern Libyan media we could find online.

The speaker distribution is heavy-tailed. A few recurring hosts dominate: the ten largest speakers hold 26% of all audio, and the Gini coefficient over speaker-hours is 0.73 (Figure 3). This is the main reason our splits must be speaker-disjoint (Section 4). The corpus is also skewed by gender: about 83% of speakers are male, consistent with a source mix of mainly TV drama, plus talk shows, podcasts, and short social-media clips; Fezzan is the most balanced, at 26% female speakers.

Region	Hours	Spk	Chan.	Ep.	Clean
Tripolitania	56.7	275	72	686	92.5
Cyrenaica	56.4	240	64	478	95.7
Fezzan	20.6	130	43	165	92.6
Total	133.7	645	161	1,281	93.9

Table 4: Corpus statistics by region; **Clean** is the percentage of segments above a 15 dB SNR threshold. Channel and episode totals are corpus-unique and are smaller than the column sums because validation can move a speaker to a region other than its channel default.

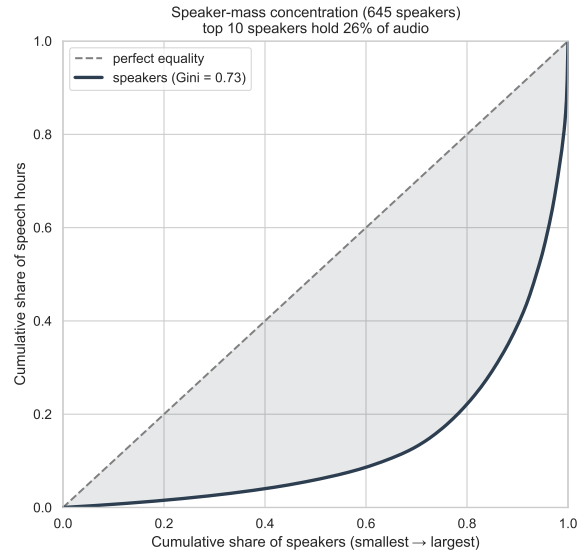


Figure 3: Lorenz curve of speaker-hours. A small number of recurring hosts hold a large share of the audio (Gini 0.73).

3.5 Release

We do not redistribute source audio from YouTube or TikTok. We release per-segment manifests with the source identifier, start and end times, region label, anonymous speaker identifier, and split assignment, plus the processing pipeline that creates the segments. For TikTok items, the manifest lists the account handle and local recording filename, so the source identifier can rebuild audio only for the YouTube segments. The release also includes cached 192-dimensional speaker-recognition ECAPA and 256-dimensional VoxLingua107 LID-ECAPA embeddings, both from pre-trained models only. It excludes embeddings from the fine-tuned E6, so no label information enters the released features. The cached embeddings let researchers reproduce the probe benchmarks even if source videos disappear. The manifests and embeddings are under CC BY 4.0 and the processing

Split	Trip.	Cyr.	Fez.	Total	Spk
train	45.4	45.1	16.5	107.0	280
dev	5.7	5.6	2.1	13.4	183
test	5.7	5.6	2.1	13.4	182

Table 5: Split composition in hours per region (rounded to 0.1 h) with distinct speaker counts. Rounded cells need not sum exactly to the totals in Table 4.

code under the MIT License. These licenses do not cover the underlying YouTube or TikTok audio, and we will remove any speaker’s entries on request.

4 Dataset Splits

We release train/dev/test splits that are speaker-disjoint and stratified by region. A speaker (after cross-episode merging) appears in exactly one split. We assign whole speakers greedily, largest first, to the split furthest below its target share, balancing by audio duration within each region. The target ratio is 80/10/10 by duration.

Speaker-disjointness matters here because of the heavy tail in Figure 3. If the same host appeared in both train and test, a model could reach high accuracy by recognizing the host rather than the dialect. Table 5 shows the result: every region is present in dev and test, and even Fezzan has about two hours and 34 speakers in each evaluation split. Dev and test carry near-identical duration but differ by a single speaker (183 vs. 182): speakers are assigned whole, so matched duration targets need not yield matched speaker counts.

5 Dialect Identification Benchmark

5.1 Motivation: off-the-shelf DID misses Libyan (E1)

Before building a sub-regional model, we checked how an existing Arabic DID system treats our data. We ran the same off-the-shelf HuBERT Arabic dialect classifier used in triage over 799 playlist-level speaker clusters, before cross-episode merging produced the 645 released speakers. For each cluster, we scored up to 10 long, clean segments, averaged the softmax outputs, and took the argmax as the cluster prediction. It labels only 74% of all clusters as Libyan (and 78% of the held-out test split), and the error rate is worst for Fezzan (68% recall). When it is wrong, it spreads the predictions across Syrian, Jordanian, Egyptian, Algerian, and Gulf labels rather than confusing Libyan with

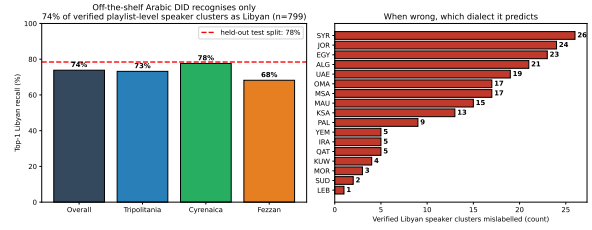


Figure 4: Off-the-shelf Arabic DID on 799 verified Libyan playlist-level speaker clusters, before cross-episode merging. Left: top-1 Libyan recall, overall and per region. Right: when wrong, which dialect the model predicts.

one neighbor (Figure 4); this pattern is consistent with documented cross-corpus degradation in Arabic DID (Shon et al., 2020; Sullivan et al., 2023). The model is confident the audio is Arabic (median Arabic probability 0.99) but does not consistently place it as Libyan. Because this classifier also fed the triage score, the 74% is a recall estimate on human-verified clusters, not an unbiased population estimate; the human, not the model, made every keep decision. The classifier also has no sub-national Libyan classes, so it cannot perform the task we define next.

5.2 Task and protocol

The task is three-way segment classification: given a speech segment, predict Tripolitania, Cyrenaica, or Fezzan. We train on the train split, select on dev, and report on test. We report accuracy and macro-averaged F1; macro-F1 is the headline metric because the regions are imbalanced and it weights each region equally. We report both segment-level results and speaker-level results, where a speaker’s prediction is the majority vote over its segments.

5.3 Models

Majority (E0). Predict the most frequent training region for everything. This is the floor. **Off-the-shelf (E1).** The country-level classifier of the previous subsection, which has no Libyan subclasses and serves as motivation, not as a sub-regional model. **ECAPA + probe (E2).** A linear classifier on the cached 192-dimensional ECAPA speaker embeddings (Desplanques et al., 2020); this reuses embeddings the pipeline already computed, so it costs no extra inference. We report both a logistic-regression and a linear-SVM head. **XLS-R + probe (E3).** A logistic-regression probe over mean-pooled hidden states from a frozen XLS-R-300m encoder (Babu et al., 2022), with

no fine-tuning. This feature is more phonetic than the speaker-oriented ECAPA embedding. **Fusion (E4)**. A logistic regression on the concatenation of the ECAPA and XLS-R features, to test whether the speaker-oriented and phonetic features carry complementary regional cues. **LID-ECAPA + probe (E5)**. The same linear probe, but over 256-dimensional embeddings from the VoxLingua107 language-identification ECAPA (Valk and Alumäe, 2021): the same ECAPA-TDNN architecture as E2, pretrained to discriminate 107 languages rather than speakers. **LID-ECAPA fine-tuned (E6)**. The VoxLingua107 ECAPA fine-tuned end to end with a fresh three-way head, the only model here that updates the encoder. To counter the Fezzan imbalance we use class-weighted cross-entropy with square-root inverse-frequency weights and SpecAugment on the input features. Full hyperparameters for every model and the significance-test procedures are in Appendix B.

5.4 Results

Table 6 reports the benchmark. The frozen language-ID ECAPA probe (E5) reaches 0.660 test macro-F1 and fine-tuning it (E6) reaches 0.702 (95% bootstrap interval [0.691, 0.714]). The encoder swap is the largest single gain: E5 exceeds the fusion E4 by +0.126 macro-F1, with a speaker-clustered bootstrap 95% CI [0.071, 0.182] and McNemar $p < 10^{-60}$. Fine-tuning from E5 to E6 adds +0.043 per segment (segment bootstrap [0.031, 0.055]; McNemar $p < 10^{-32}$): E6 corrects 1,217 test segments that E5 got wrong and turns 695 correct E5 predictions into errors. This gain is not significant under the speaker-clustered bootstrap, CI [-0.011, 0.093]. E5 and E6 reach about the same Fezzan F1, so the E6 improvement lies in the two coastal regions. At speaker level, E5 leads E6 on dev and E6 leads E5 on test (Table 6). Model selection was fixed in advance on dev segment-level macro-F1; speaker-level voting was computed afterwards and never used for selection. For speaker-level prediction, the frozen E5 probe reaches 0.734 test macro-F1 against 0.750 for E6, without any end-to-end fine-tuning.

The per-class numbers in Table 7 show which regions account for the gains. E3 has the highest Fezzan precision of any model (0.87), yet it almost never predicts Fezzan, so recall collapses to 0.066 and F1 to 0.12. E6 improves every region, with Fezzan F1 rising from 0.36 in E2 to 0.53. Most of the Fezzan gain comes from the encoder: the

Model	Lvl	Dev		Test	
		Acc	F1	Acc	F1
Majority (E0)	seg	43.1	20.1	43.2	20.1
ECAPA+logreg (E2)	seg	55.3	50.4	57.4	52.5
ECAPA+SVM (E2)	seg	55.8	50.6	58.0	52.7
XLS-R+probe (E3)	seg	52.8	42.2	56.6	44.4
Fusion (E4)	seg	56.2	51.2	58.4	53.4
LID-ECAPA+probe (E5)	seg	68.3	64.7	69.0	66.0
LID-ECAPA-FT (E6)	seg	73.6	68.6	75.5	70.2
ECAPA+logreg (E2)	spk	54.6	51.1	57.1	50.5
XLS-R+probe (E3)	spk	57.4	45.1	59.3	46.6
Fusion (E4)	spk	57.4	54.0	57.7	51.7
LID-ECAPA+probe (E5)	spk	77.1	75.2	78.0	73.4
LID-ECAPA-FT (E6)	spk	74.9	69.2	79.7	75.0

Table 6: Three-way sub-regional DID (%). Macro-F1 is the headline metric; best test F1 per level in bold. Model E1 (the off-the-shelf country-level classifier, §5) has no sub-regional classes and cannot produce three-way predictions, so it is omitted here. The E5→E6 macro-F1 gain has a speaker-clustered 95% CI of [-0.011, 0.093].

frozen LID probe E5 already reaches Fezzan F1 0.55, marginally above E6. End-to-end fine-tuning with class weighting sharpens the two data-rich coastal regions at a small cost to Fezzan recall (0.45). Fezzan remains the hardest class, but E5 and E6 predict it far more often than E3 does.

5.5 Analysis

The confusion matrix for the fine-tuned model (Figure 5) shows where the difficulty remains. Fezzan is the main source of error: even fine-tuned it reaches only 45% recall, and its missed segments go mostly to Tripolitania (41%) rather than Cyrenaica (15%). Fezzan has the least training data (16.5 hours), which lowers its recall. Class size does not explain this direction: the two coastal regions have almost the same training hours (45.4 and 45.1). One hypothesis is that formal broadcast settings lead speakers, many of them educated, to shift toward MSA and accommodate toward the Tripoli variety by softening southern features. The segment audit points in the same direction as the accommodation hypothesis: 33.3% of Fezzan test segments were judged MSA. Among dialectal Fezzan test segments, the first annotator heard Tripolitania as often as Fezzan (12 segments each, versus 6 Cyrenaica); for no other released region did the annotator choose a neighbouring region this often. Genuine dialect closeness remains a possible cause, and neither the model nor the audit can separate the two effects. Corpus composition and the small

Region	spk-ECAPA (E2)			XLS-R (E3)			LID-ECAPA frozen (E5)			LID-ECAPA-FT (E6)		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Tripolitania	.598	.565	.581	.542	.673	.601	.671	.728	.699	.705	.888	.786
Cyrenaica	.610	.660	.634	.586	.635	.609	.746	.709	.727	.855	.733	.789
Fezzan	.377	.345	.361	.868	.066	.123	.582	.527	.553	.655	.448	.532
Macro	.528	.524	.525	.665	.458	.444	.666	.655	.660	.738	.690	.702

Table 7: Per-class precision, recall, and F1 on the test split (segment level) for models E2, E3, E5, and E6. Macro rows are averaged at full precision and may differ from the rounded per-class mean by about 0.001.

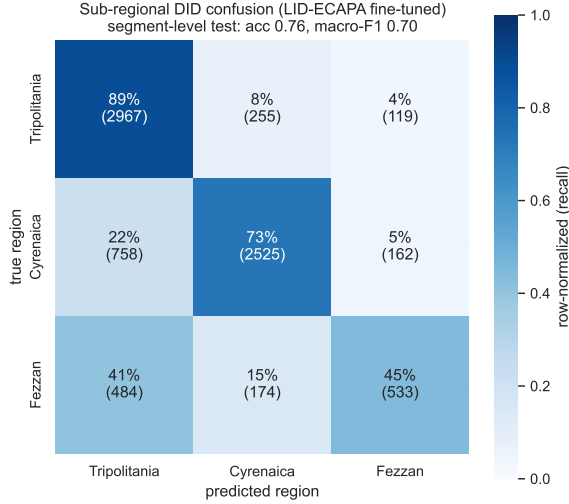


Figure 5: Test-set confusion for the fine-tuned LID-ECAPA model (E6), row-normalized recall. Fezzan is the hardest region and is confused far more often with Tripolitania (41%) than with Cyrenaica (15%).

number of Fezzan channels may also contribute.

Figure 6 plots all 645 speaker centroids from the fine-tuned E6 model on two LDA axes. Cyrenaica and Tripolitania lie on opposite sides, and Fezzan lies between them, closer to the Tripolitania side. The axes were fit only on training segments and their region labels. The 365 held-out dev and test speakers, never used to fit the axes, still separate on them with a silhouette score of 0.29, so the pattern generalizes rather than being an artefact of fitting the axes. An unsupervised t-SNE of the same embeddings gives a silhouette of 0.05: the separation is a specific direction in the embedding space that the region-supervised LDA fit has to find, not an obvious clustering. The frozen encoders show this cue more weakly. Part of what any probe reads from these embeddings may be channel- or speaker-correlated rather than dialectal, which Section 5.7 tests.

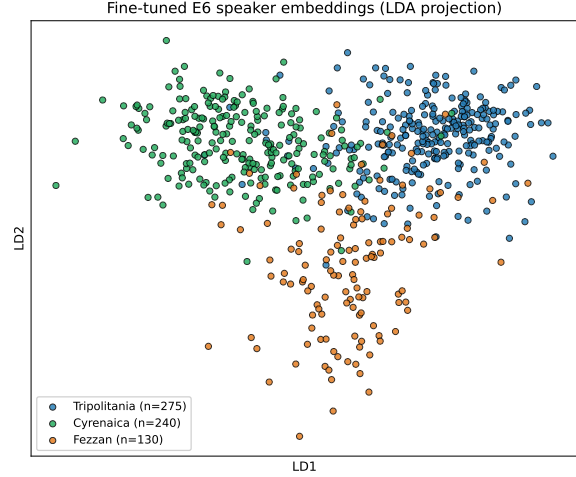


Figure 6: All 645 speaker centroids from the fine-tuned E6 embeddings, colored by region. The two LDA axes were fit on training segments only.

5.6 Robustness to segment length

Accuracy changes little across the three duration bands. The ECAPA probe (E2) scores 58.3% on three-to-five-second segments and 55.7% on the longest, and the fusion (E4) behaves the same way (59.9% down to 56.1%). The fine-tuned model (E6) rises only slightly with length, from 74.5% on 3–5 s segments to 75.3% on 5–7 s and 76.9% on 7–10 s. Segment length is therefore not a main source of error for any of these models.

5.7 Robustness to channel overlap

The released splits are speaker-disjoint: each merged speaker sits in one split. They are not channel-disjoint. A channel (playlist) can place its speakers in different splits, so 6,135 of 7,977 test segments (77%) and 153 of 182 test speakers (84%) share a channel with training, which lets a classifier read channel or recording cues that happen to track region rather than dialect.

To bound this effect we built a second split that keeps whole channels together. Each of the 161 channels is assigned by its dominant region (the

Region	ECAPA (E2)		Fusion (E4)		LID-FT (E6)	
	Spk	Chan	Spk	Chan	Spk	Chan
Tripolitania	.581	.580	.593	.579	.786	.793
Cyrenaica	.634	.591	.645	.584	.789	.722
Fezzan	.361	.225	.363	.229	.532	.313
Macro	.525	.465	.534	.464	.702	.609

Table 8: Per-region and macro F1 (test, segment level) under the speaker-disjoint (Spk) and channel-disjoint (Chan) splits for ECAPA (E2), Fusion (E4), and LID-FT (E6).

region holding most of its duration), and within a region channels are placed largest-first into train, dev, or test to reach an 80/10/10 duration share. No channel crosses splits. By duration, TikTok accounts for about 28% of the test and dev splits, compared with about 7% of training, because the largest YouTube channels were packed into train first. We then re-fit the ECAPA probe (E2) on the same cached features and retrain the fine-tuned model (E6) from scratch on this split (end-to-end fine-tuning cannot reuse cached features), and evaluate both on the channel-disjoint test set, which has 48 channels by dominant-region counting (19 Tripolitania, 17 Cyrenaica, and 12 Fezzan), or 50 when each channel is counted for every region whose speakers it carries; Fezzan has 12 under either count.

Macro-F1 falls from 0.702 to 0.609 for the fine-tuned model when no test channel is seen in training (Table 8). The drop suggests that part of the speaker-disjoint result depends on channel-correlated cues, although the channel-disjoint split also changes the source mix. The two coastal regions move little while most of the drop falls on Fezzan, which is covered by the fewest channels, so its channel-disjoint estimate rests on few independent sources and is the noisiest. For E6, 95% channel-clustered bootstrap CIs, resampling whole channels, are 0.609 [.486, .677] for macro, .793 [.606, .893] for Tripolitania, .722 [.435, .836] for Cyrenaica, and .313 [.051, .455] for Fezzan.

The channel-disjoint Fezzan F1 of 0.31, with an interval reaching down to 0.05, shows how much room for improvement remains.

6 Discussion

The front-end’s pretraining objective matters more than the classifier or the fusion: under the same linear probe, swapping the speaker-recognition

ECAPA for a language-identification ECAPA is the single largest gain, and fine-tuning that encoder adds less. A speaker encoder is trained to separate voices; a language-ID encoder is trained to separate languages, which is closer to the dialect task. The second finding is that fine-tuning sharpens the data-rich coastal regions but does not improve Fezzan over the frozen language-ID probe (E5).

These regional results are consistent with earlier descriptions of Libyan Arabic. The two coastal varieties, Cyrenaica and Tripolitania, are the easiest for every model and have the most data and the best-described, most distinct varieties (Benkato, 2014; Pereira, 2010); southern Libyan sits in a contact zone with both coastal varieties and with languages to the south, so a soft Fezzan boundary is expected. The segment audit (Section 5, Table 3) points the same way: Fezzan test segments were judged MSA more than twice as often as coastal ones, and dialectal Fezzan segments were heard as Tripolitania as often as Fezzan (12 segments each). Register and accommodation, as well as genuine dialect closeness, could produce this pattern, and the benchmark cannot separate them. Limited Fezzan training data adds error on top of this pattern but does not explain its direction. The results support the three-region partition, but the Fezzan and channel-disjoint numbers show that the evidence is still limited.

7 Conclusion

We presented a sub-regional Libyan Arabic speech corpus—133.7 hours, 645 speakers, 161 channels, three regions, speaker-disjoint splits—built by an automated pipeline and human review that confirmed the provenance region for 95.5% of kept speakers (within-annotator consistency). An off-the-shelf Arabic model recognizes Libyan only 74% of the time and cannot resolve the region. On the new three-way task, the front-end’s pre-training objective had the largest effect of any choice tested: a language-identification ECAPA reaches 0.702 macro-F1 speaker-disjoint and 0.609 channel-disjoint, both above a speaker-recognition ECAPA (0.525) and the 0.20 floor. Fezzan remains the hardest region: it has the least training data (16.5 hours) and the lowest F1, falling from 0.532 speaker-disjoint to 0.313 channel-disjoint. In the segment audit, 33.3% of Fezzan test segments were judged MSA. Closing this gap needs more dialectal southern speech from more channels.

Limitations

The regional labels are provenance-based: they start from the origin of the source channel, not from per-utterance perceptual annotation by multiple linguists. A guest from one region appearing on another region’s channel is a source of label noise that our human review only partly corrects. The 95.5% rate is within-annotator consistency; the blind review (Section 3.3) gives the inter-annotator figure. That review was done by a co-author. It measures agreement between the two annotators; it does not measure anchoring, which we did not quantify. Validation is also speaker-level, not segment-level: the annotator made one keep-or-relabel decision per speaker after listening to a spread of that speaker’s clips, but did not audit every segment individually.

Automated gates within each speaker—an embedding-cohesion audit and the music, voice-activity, and signal-to-noise filters—remove off-cluster and non-speech segments, but segments judged MSA can still pass under the speaker’s label: 17.3% of test segments overall and 33.3% of Fezzan test segments (Table 3); code-switched and mixed-speaker segments were rare.

Several of the regions we label also contain Berber/Amazigh-speaking communities (for example the Nafusa Mountains and Zuwarah in the west, Awjila in the east, and the Tuareg area of the south (Benkato and Pereira, 2016; *Encyclopædia Britannica*, 2025)), whose speech is not Arabic. Our human review removes obvious non-Libyan and MSA speakers, but we apply no explicit Arabic-versus-Berber gate, so a small amount of Berber or Berber-influenced speech may remain under a regional Arabic label.

The corpus is broadcast and social-media speech and is skewed toward male speakers (83%), so results may not transfer to conversational or read speech, or to balanced-gender settings. Fezzan is the smallest class (20.6 hours; 165 episodes, compared with 686 for Tripolitania and 478 for Cyrenaica), and we found little more usable Fezzan material after established broadcast sources ran out. Given the MSA rate in the segment audit, the Fezzan data may not represent everyday southern speech.

All our encoders, including the fine-tuned language-identification ECAPA, encode speaker and channel as well as dialect, and even after fine-tuning the regions only partly separate in Figure 6. The channel-disjoint re-evaluation (Section 5.7) in-

dicates that part of the signal is channel-correlated: the fine-tuned model’s macro-F1 falls from 0.702 to 0.609 and Fezzan’s F1 from 0.53 to 0.31 once no test channel is seen in training. With every method tested here, Fezzan recall stays below 0.45 speaker-disjoint and below 0.21 channel-disjoint; other imbalance-aware methods remain to be evaluated.

We did not test end-to-end fine-tuning of the self-supervised encoder or larger language-identification models. We also report no per-split genre breakdown, because genre is not in our metadata, and no diarization error rate, because we have no time-aligned reference.

Ethics Statement

The audio is public Libyan media collected for research. We assign each speaker an anonymous ID and attach no names. For TikTok records, the manifest keeps the public account handle of the publisher, and because most TikTok accounts contain a single speaker, this handle identifies the speaker directly. We release no other personal data about account holders or speakers. We do not redistribute the source audio; we release segment manifests, derived embeddings, and code, consistent with the source platforms’ terms. The corpus is intended for research on Arabic dialect variation and speech processing. Regional dialect labels are sensitive: they describe recorded media, not claims about any individual’s identity or origin.

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A Speaker Counts by Validation Stage

Stage	Count
Candidate speaker identities above the evidence floor, all reviewed by the annotator	769
Identities kept after the annotator confirmed them as Libyan (76 excluded: 34 MSA, 28 poor audio, 13 non-Libyan, 1 mixed)	693
The same 693 confirmed speakers counted as playlist-level speaker clusters, before cross-playlist merging; the unit of Section 5.1 and Figure 4	799
Final speakers after the annotator manually merged identities across playlists	645

Table 9: Speaker counts at validation stages.

B Implementation Details

Pretrained checkpoints. Diarization: pyannote/speaker-diarization-community-1. Voice activity: Silero VAD (default thresholds). Music scoring: PANNs CNN14. Speaker embeddings: speechbrain/spkrec-ecapa-voxceleb (192-d). Language-ID embeddings and the E5/E6 encoder: speechbrain/lang-id-voxlinaua107-ecapa (256-d). Self-supervised features: facebook/wav2vec2-xls-r-300m. Triage and E1 classifier: IbrahimAmin/hubert-arabic-spoken-dialect-classifier. All checkpoints come from the Hugging Face Hub except Silero and PANNs, which come from their GitHub releases.

Frozen-feature probes (E2–E5). All probes are scikit-learn (Pedregosa et al., 2011) logistic regression (L2, $C=1$, 5000 iterations) or linear SVM on L2-normalized features. The XLS-R feature is the masked mean of the frozen encoder’s last hidden state, capped at 15 s (1024-d); the fusion concatenates the two L2-normalized vectors. Split seed 1337.

Fine-tuning (E6). E6 fine-tunes the ECAPA-TDNN encoder and a linear head (about 21M parameters) with AdamW (encoder 10^{-4} , head 10^{-3}), batch 16, bf16, linear warmup then decay, and early stopping on dev macro-F1; SpecAugment masks time and frequency on the Fbank features in training only. Split seed 1337.

Significance testing. All confidence intervals are percentile bootstraps over the test segments (1,000 resamples, seed 1337). The channel-disjoint intervals instead resample whole channels (1,000 resamples, seed 1337). Because segments from one speaker are correlated, this i.i.d. segment resampling makes the intervals mildly optimistic; CI width is driven mainly by the test-set size, so models with very different macro-F1 can have similar widths. Paired comparisons add McNemar’s exact test on segment correctness and a paired bootstrap of the macro-F1 difference, run both i.i.d. over segments and clustered by speaker (resampling whole speakers) to account for that correlation.